Editor

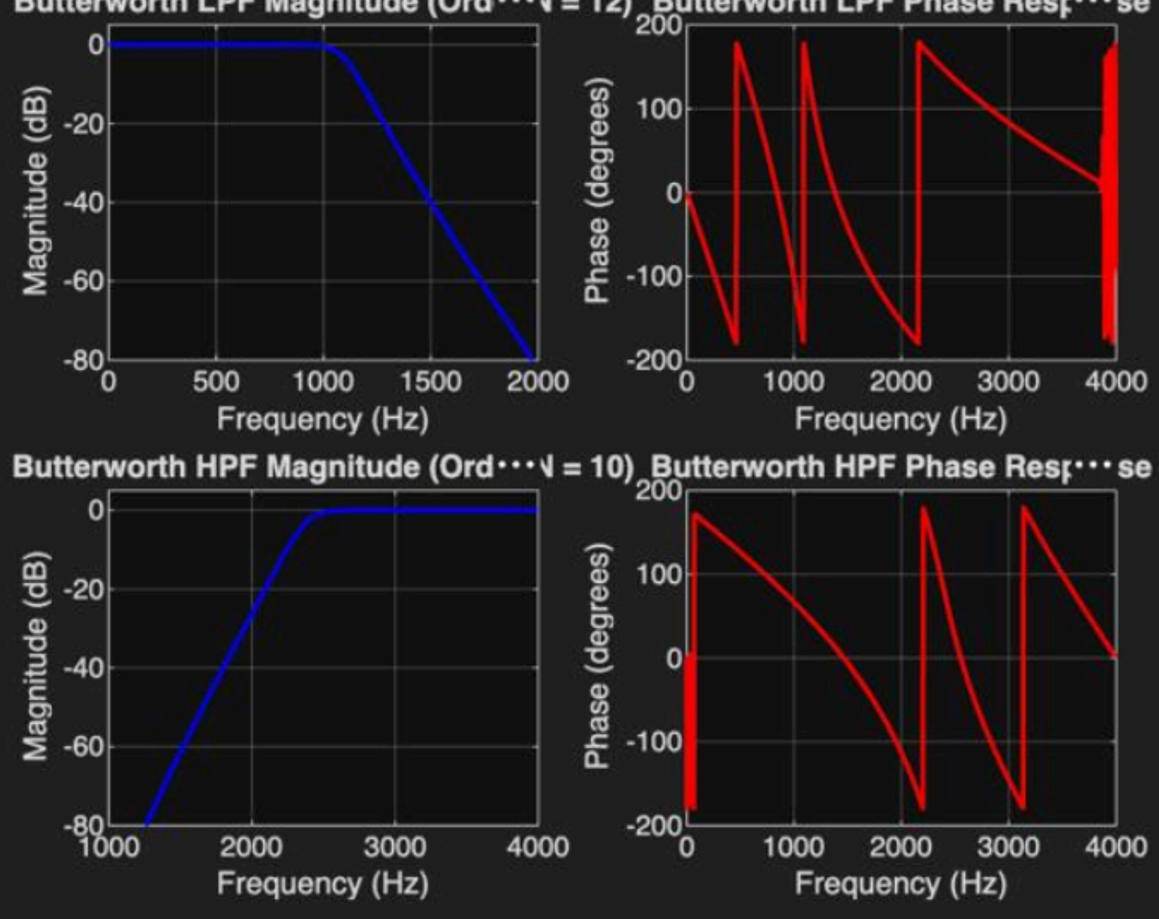
i5g.m × | exp115g6g.m × | exp125g6g.m × | exp135g6g.m × | exp5g6g.m × | exp155g6g.m × | smart.m × | untitled.m × | ML.m × | ML2.m × | mlll.m × | mageshco1exp1.m ×

/MATLAB Drive/mageshco1exp1.m

```
clear;
clc;
close all;
% Design Parameters
Fs = 8000; Ap = 1; As = 40; % Sampling frequency in Hz
% Passband ripple in dB
% Stopband attenuation in dB
% 1. Low-Pass Filter Design
fp_lp = 1000; fs_lp = 1500; wp_lp = fp_lp / (Fs/2);
ws_lp = fs_lp / (Fs/2);
% LPF Passband frequency (Hz)
% LPF Stopband frequency (Hz)
[N_lp, Wn_lp] = buttord(wp_lp, ws_lp, Ap, As);
[b_lp, a_lp] = butter(N_lp, Wn_lp, 'low');
% 2. High-Pass Filter Design
fp_hp = 2500; fs_hp = 1800; wp_hp = fp_hp / (Fs/2);
ws_hp = fs_hp / (Fs/2);
% HPF Passband frequency (Hz)
% HPF Stopband frequency (Hz)
[N_hp, Wn_hp] = buttord(wp_hp, ws_hp, Ap, As);
[b_hp, a_hp] = butter(N_hp, Wn_hp, 'high');
% ========================================
% DISPLAY NUMERICAL OUTPUTS IN COMMAND WINDOW
% ========================================
fprintf('=== EXPERIMENT 1: BUTTERWORTH IIR FILTERS ===\n\n');
fprintf('--- LOW-PASS FILTER ---\n');
fprintf('Calculated Filter Order (N_lp): %d\n', N_lp);
fprintf('3-dB Cutoff Frequency (Wn_lp): %.4f Hz\n', Wn_lp * (Fs/2));
fprintf('Numerator Coefficients (b_lp):\n');
disp(b_lp);
fprintf('Denominator Coefficients (a_lp):\n');
disp(a_lp);
```

Figure 1: Butterworth IIR LPF and HPF

FIGURE FORMAT TOOLS LAYOUT



Command Window

New to MATLAB? See resources for Getting Started.

```
3-dB Cutoff Frequency (Wn_hp): 2379.7556 Hz
Numerator Coefficients (b_hp):
    0.0006   -0.0055    0.0248   -0.0662    0.1159   -0.1390    0.1159   -0.0662    0.0248   -0.0055    0.0006

Denominator Coefficients (a_hp):
    1.0000    1.8915    2.8535    2.6014    1.8704    0.9534    0.3726    0.1030    0.0199    0.0023    0.0001

>>
```